Pandas 101: One-stop Shop for Data Science

This notebook can be treated as pandas cheatsheet or a beginner-friendly guide to learn from basics.

Last updated on 24-May-2020 (Appending & Concatenating Series)

8. Detecting missing values .isna()

9. Counting missing values

10. Removing missing values

11. Adding a new column

12. Deleting columns in DataFrame

13. Summary statistics

14. agg() method

15. Dropping duplicate names

16. Count categorical data

17. Grouped summaries

18. Pivot table

19. Explicit indexes

20. Visualizing your data

21. Arithmetic with Series & DataFrames

21. Merge DataFrames

23. What next?

"Avocado Prices" dataset is used in this notebook :)

In [3]:
```
import numpy as np
import pandas as pd
%matplotlib inline
import matplotlib.pyplot as plt
```

Creating DataFrames

* From a list of dictionaries (constructed row by row)

In [5]:
```
list_of_dicts = [
     {"name": "Ginger", "breed": "Dachshund", "height_cm": 22,"weight_kg": 10, "date_of_birth": "2019-03-14"},
    {"name": "Scout", "breed": "Dalmatian", "height_cm": 59,"weight_kg": 25, "date_of_birth": "2019-05-09"}
```

```
]
new_dogs = pd.DataFrame(list_of_dicts)
new_dogs
```

Out[5]:

	name	breed	height_cm	weight_kg	date_of_birth
0	Ginger	Dachshund	22	10	2019-03-14
1	Scout	Dalmatian	59	25	2019-05-09

* From a dictionary of lists (constructed column by column)

In [6]:
```
dict_of_lists = {
     "name": ["Ginger", "Scout"],
     "breed": ["Dachshund", "Dalmatian"],
     "height_cm": [22, 59],
     "weight_kg": [10, 25],
     "date_of_birth": ["2019-03-14","2019-05-09"]  }
new_dogs = pd.DataFrame(dict_of_lists)
new_dogs
```

Out[6]:

	name	breed	height_cm	weight_kg	date_of_birth
0	Ginger	Dachshund	22	10	2019-03-14
1	Scout	Dalmatian	59	25	2019-05-09

Reading and writing CSVs

CSV = comma-separated values

Designed for DataFrame-like data

Most database and spreadsheet programs can use them or create them

In [7]:
```
# read CSV from using pandas
avocado = pd.read_csv(r"D:\Samson - All Data\Naresh IT Institute\New folder\avocado.csv")
# print the first few rows of the dataframe
avocado.head()
```

Out[7]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	Albany
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	Albany
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	Albany
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	Albany
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	Albany

Read CSV and assign index

You can assign columns as index using "index_col" attribute.

Since I want to index Date there is another helpful function called "parse_date" which will parse the date in the rows such that we can perform more complex subsetting(eg monthly, weekly etc).

In [8]:
```
# read CSV from using pandas and assigning Date as index of the dataframe
avocado = pd.read_csv(r"D:\Samson - All Data\Naresh IT Institute\New folder\avocado.csv",parse_dates=True, index_col='Date')
# print the first few rows of the dataframe
avocado.head()
```

Out[8]:

	Unnamed: 0	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
Date													
2015-12-27	0	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	Albany
2015-12-20	1	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	Albany
2015-12-13	2	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	Albany
2015-12-06	3	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	Albany
2015-11-29	4	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	Albany

Remove index from dataframe .reset_index(drop)

To reset the index use this function

In [9]:
```
avocado = avocado.reset_index(drop=True)
avocado.head()
```

Out[9]:

	Unnamed: 0	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
0	0	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	Albany
1	1	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	Albany
2	2	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	Albany
3	3	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	Albany
4	4	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	Albany

In [11]:
```
#To write a CSV file function dataframe.to_csv(FILE_NAME)
```

In [12]:
```
avocado.to_csv("test_write.csv")
```

Some useful pandas function

* **.head()** or **.head(x)** is used to get the first x rows of the DataFrame (x = 5 by default)

In [14]:
```
avocado = pd.read_csv(r"D:\Samson - All Data\Naresh IT Institute\New folder\avocado.csv")
avocado.head()
```

Out[14]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	Albany
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	Albany
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	Albany
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	Albany
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	Albany

* **.tail()** or **.tail(x)** is used to get the last x rows of the DataFrame (x = 5 by default)

In [15]:
```
avocado.tail(10)
```

Out[15]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
18239	2	2018-03-11	1.56	22128.42	2162.67	3194.25	8.93	16762.57	16510.32	252.25	0.0	organic	2018	WestTexNew
18240	3	2018-03-04	1.54	17393.30	1832.24	1905.57	0.00	13655.49	13401.93	253.56	0.0	organic	2018	WestTexNew
18241	4	2018-02-25	1.57	18421.24	1974.26	2482.65	0.00	13964.33	13698.27	266.06	0.0	organic	2018	WestTexNew
18242	5	2018-02-18	1.56	17597.12	1892.05	1928.36	0.00	13776.71	13553.53	223.18	0.0	organic	2018	WestTexNew
18243	6	2018-02-11	1.57	15986.17	1924.28	1368.32	0.00	12693.57	12437.35	256.22	0.0	organic	2018	WestTexNew
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	organic	2018	WestTexNew
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	organic	2018	WestTexNew
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	WestTexNew
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	WestTexNew
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	WestTexNew

* .info() is used to get a concise summary of the DataFrame

In [16]:

```
avocado.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18249 entries, 0 to 18248
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   Unnamed: 0    18249 non-null  int64  
 1   Date          18249 non-null  object 
 2   AveragePrice  18249 non-null  float64
 3   Total Volume  18249 non-null  float64
 4   4046          18249 non-null  float64
 5   4225          18249 non-null  float64
 6   4770          18249 non-null  float64
 7   Total Bags    18249 non-null  float64
 8   Small Bags    18249 non-null  float64
 9   Large Bags    18249 non-null  float64
 10  XLarge Bags   18249 non-null  float64
 11  type          18249 non-null  object 
 12  year          18249 non-null  int64  
 13  region        18249 non-null  object 
dtypes: float64(9), int64(2), object(3)
memory usage: 1.9+ MB
```

* **.shape** is used to get the dimensions of the DataFrame

In [17]:
```
print(avocado.shape)
```

```
(18249, 14)
```

* **.describe()** is used to view some basic statistical details like percentile, mean, std etc. of a DataFrame

In [18]:
```
avocado.describe()
```

Out[18]:

	Unnamed: 0	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags
count	18249.000000	18249.000000	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04
mean	24.232232	1.405978	8.506440e+05	2.930084e+05	2.951546e+05	2.283974e+04	2.396392e+05	1.821947e+05	5.433809e+04
std	15.481045	0.402677	3.453545e+06	1.264989e+06	1.204120e+06	1.074641e+05	9.862424e+05	7.461785e+05	2.439660e+05
min	0.000000	0.440000	8.456000e+01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
25%	10.000000	1.100000	1.083858e+04	8.540700e+02	3.008780e+03	0.000000e+00	5.088640e+03	2.849420e+03	1.274700e+02
50%	24.000000	1.370000	1.073768e+05	8.645300e+03	2.906102e+04	1.849900e+02	3.974383e+04	2.636282e+04	2.647710e+03
75%	38.000000	1.660000	4.329623e+05	1.110202e+05	1.502069e+05	6.243420e+03	1.107834e+05	8.333767e+04	2.202925e+04
max	52.000000	3.250000	6.250565e+07	2.274362e+07	2.047057e+07	2.546439e+06	1.937313e+07	1.338459e+07	5.719097e+06

* **.values** this attribute return a Numpy representation of the given DataFrame

In [19]:

```
avocado.values
```

Out[19]:

```
array([[0, '2015-12-27', 1.33, ..., 'conventional', 2015, 'Albany'],
       [1, '2015-12-20', 1.35, ..., 'conventional', 2015, 'Albany'],
       [2, '2015-12-13', 0.93, ..., 'conventional', 2015, 'Albany'],
       ...,
       [9, '2018-01-21', 1.87, ..., 'organic', 2018, 'WestTexNewMexico'],
       [10, '2018-01-14', 1.93, ..., 'organic', 2018, 'WestTexNewMexico'],
       [11, '2018-01-07', 1.62, ..., 'organic', 2018, 'WestTexNewMexico']],
      shape=(18249, 14), dtype=object)
```

* **.columns** this attribute return a Numpy representation of columns in the DataFrame

In [20]:
```
print(avocado.columns)
```

```
Index(['Unnamed: 0', 'Date', 'AveragePrice', 'Total Volume', '4046', '4225',
       '4770', 'Total Bags', 'Small Bags', 'Large Bags', 'XLarge Bags', 'type',
       'year', 'region'],
      dtype='object')
```

Appending & Concatenating Series

append(): Series & DataFrame method

- Invocation:
- s1.append(s2)
- Stacks rows of s2 below s1

concat(): pandas module function

- Invocation:
- pd.concat([s1, s2, s3])
- Can stack row-wise or column-wise

In [23]:
```
even = pd.Series([2,4,6,8,10])
odd = pd.Series([1,3,5,7,9])

res = pd.concat([even, odd], ignore_index=True)
print(res)
```

```
0     2
1     4
2     6
3     8
4    10
5     1
6     3
7     5
8     7
9     9
dtype: int64
```

Observe index got messed up

You can use .reset_index(drop=True) to fix it

Note: if drop = False then previous index will be added as a column

In [24]:
```
res.reset_index(drop=True)
```

Out[24]:
```
0     2
1     4
2     6
3     8
4    10
5     1
6     3
7     5
8     7
9     9
dtype: int64
```

Sorting

syntax:

> DataFrame.sort_values(by, axis=0, ascending=True, inplace=False, kind='quicksort', na_position='last')

- by: Single/List of column names to sort Data Frame by.
- axis: 0 or ‘index’ for rows and 1 or ‘columns’ for Column.
- ascending: Boolean value which sorts Data frame in ascending order if True.
- inplace: Boolean value. Makes the changes in passed data frame itself if True.
- kind: String which can have three inputs(‘quicksort’, ‘mergesort’ or ‘heapsort’) of algorithm used to sort data frame.
- na_position: Takes two string input ‘last’ or ‘first’ to set position of Null values. Default is ‘last’.

In [25]:
```
# sort values based on "AveragePrice" (ascending) and "year" (descending)
avocado.sort_values(["AveragePrice", "year"], ascending=[True, False])
```

Out[25]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	typ
15261	43	2017-03-05	0.44	64057.04	223.84	4748.88	0.00	59084.32	638.68	58445.64	0.00	organ
7412	47	2017-02-05	0.46	2200550.27	1200632.86	531226.65	18324.93	450365.83	113752.17	330583.10	6030.56	convention
15473	43	2017-03-05	0.48	50890.73	717.57	4138.84	0.00	46034.32	1385.06	44649.26	0.00	organ
15262	44	2017-02-26	0.49	44024.03	252.79	4472.68	0.00	39298.56	600.00	38698.56	0.00	organ
1716	0	2015-12-27	0.49	1137707.43	738314.80	286858.37	11642.46	100891.80	70749.02	30142.78	0.00	convention
...	...	...	...	...	...	...	...	...	...	...	...	
16720	18	2017-08-27	3.04	12656.32	419.06	4851.90	145.09	7240.27	6960.97	279.30	0.00	organ
16055	42	2017-03-12	3.05	2068.26	1043.83	77.36	0.00	947.07	926.67	20.40	0.00	organ
14124	7	2016-11-06	3.12	19043.80	5898.49	10039.34	0.00	3105.97	3079.30	26.67	0.00	organ
17428	37	2017-04-16	3.17	3018.56	1255.55	82.31	0.00	1680.70	1542.22	138.48	0.00	organ
14125	8	2016-10-30	3.25	16700.94	2325.93	11142.85	0.00	3232.16	3232.16	0.00	0.00	organ

18249 rows × 14 columns

Sorting by index

use df.sort_index(ascending=True/False)

Subsetting

Subsetting is used to get a slice of the original dataframe

In [27]:
```
# Subsetting columns
avocado["AveragePrice"]
```

Out[27]:
```
0        1.33
1        1.35
2        0.93
3        1.08
4        1.28
         ...
18244    1.63
18245    1.71
18246    1.87
18247    1.93
18248    1.62
Name: AveragePrice, Length: 18249, dtype: float64
```

Subsetting multiple columns

In [28]:
```
# Subsetting multiple columns
avocado[["AveragePrice","Date"]]
```

Out[28]:

	AveragePrice	Date
0	1.33	2015-12-27
1	1.35	2015-12-20
2	0.93	2015-12-13
3	1.08	2015-12-06
4	1.28	2015-11-29
...	...	...
18244	1.63	2018-02-04
18245	1.71	2018-01-28
18246	1.87	2018-01-21
18247	1.93	2018-01-14
18248	1.62	2018-01-07

18249 rows × 2 columns

Subsetting rows

In [29]:
```
# Subsetting rows
avocado["AveragePrice"]<1
```

Out[29]:

```
0        False
1        False
2         True
3        False
4        False
         ...  
18244    False
18245    False
18246    False
18247    False
18248    False
Name: AveragePrice, Length: 18249, dtype: bool
```

and then using it for subsetting the original dataframe

In [30]:

```python
# This will print only the rows with price < 1
avocado[avocado["AveragePrice"]<1]
```

Out[30]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	ye
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	20
6	6	2015-11-15	0.99	83453.76	1368.92	73672.72	93.26	8318.86	8196.81	122.05	0.0	conventional	20
7	7	2015-11-08	0.98	109428.33	703.75	101815.36	80.00	6829.22	6266.85	562.37	0.0	conventional	20
13	13	2015-09-27	0.99	106803.39	1204.88	99409.21	154.84	6034.46	5888.87	145.59	0.0	conventional	20
43	43	2015-03-01	0.99	55595.74	629.46	45633.34	181.49	9151.45	8986.06	165.39	0.0	conventional	20
...	...	...	...	...	...	...	...	...	...	...	...	...	
17169	43	2017-03-05	0.99	155011.12	35367.23	5175.81	5.91	114462.17	95379.07	19083.10	0.0	organic	20
17170	44	2017-02-26	0.99	171145.00	34520.03	6936.39	0.00	129688.58	117252.31	12436.27	0.0	organic	20
17536	39	2017-04-02	0.98	402676.23	34093.33	58330.53	207.85	310044.52	155701.41	154343.11	0.0	organic	20
17537	40	2017-03-26	0.90	456645.91	36169.35	51398.72	139.55	368938.29	152159.53	216778.76	0.0	organic	20
17540	43	2017-03-05	0.99	367519.17	61166.48	55123.99	126.80	251101.90	112844.19	138257.71	0.0	organic	20

2796 rows × 14 columns

Subsetting based on text data

In [31]:
```python
# it will print all the rows with "type" = "organic"
avocado[avocado["type"]=="organic"]
```

Out[31]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
9126	0	2015-12-27	1.83	989.55	8.16	88.59	0.00	892.80	892.80	0.00	0.0	organic	2015	
9127	1	2015-12-20	1.89	1163.03	30.24	172.14	0.00	960.65	960.65	0.00	0.0	organic	2015	
9128	2	2015-12-13	1.85	995.96	10.44	178.70	0.00	806.82	806.82	0.00	0.0	organic	2015	
9129	3	2015-12-06	1.84	1158.42	90.29	104.18	0.00	963.95	948.52	15.43	0.0	organic	2015	
9130	4	2015-11-29	1.94	831.69	0.00	94.73	0.00	736.96	736.96	0.00	0.0	organic	2015	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	organic	2018	WestTexNew
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	organic	2018	WestTexNew
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	WestTexNew
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	WestTexNew
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	WestTexNew

9123 rows × 14 columns

Subsetting based on dates

In [32]:
```
# it will print all the rows with "Date" <= 2015-02-04
avocado[avocado["Date"]<="2015-02-04"]
```

Out[32]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
47	47	2015-02-01	0.99	70873.60	1353.90	60017.20	179.32	9323.18	9170.82	152.36	0.0	conventional	2015	
48	48	2015-01-25	1.06	45147.50	941.38	33196.16	164.14	10845.82	10103.35	742.47	0.0	conventional	2015	
49	49	2015-01-18	1.17	44511.28	914.14	31540.32	135.77	11921.05	11651.09	269.96	0.0	conventional	2015	
50	50	2015-01-11	1.24	41195.08	1002.85	31640.34	127.12	8424.77	8036.04	388.73	0.0	conventional	2015	
51	51	2015-01-04	1.22	40873.28	2819.50	28287.42	49.90	9716.46	9186.93	529.53	0.0	conventional	2015	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
11928	46	2015-02-01	1.77	7210.19	1634.42	3012.44	0.00	2563.33	2563.33	0.00	0.0	organic	2015	West
11929	47	2015-01-25	1.63	7324.06	1934.46	3032.72	0.00	2356.88	2320.00	36.88	0.0	organic	2015	West
11930	48	2015-01-18	1.71	5508.20	1793.64	2078.72	0.00	1635.84	1620.00	15.84	0.0	organic	2015	West
11931	49	2015-01-11	1.69	6861.73	1822.28	2377.54	0.00	2661.91	2656.66	5.25	0.0	organic	2015	West
11932	50	2015-01-04	1.64	6182.81	1561.30	2958.17	0.00	1663.34	1663.34	0.00	0.0	organic	2015	West

540 rows × 14 columns

Subsetting based on multiple conditions

You can use the logical operators to define a complex condition

- "&" and
- "|" or
- "~" not

> ** SEPERATE EACH CONDITION WITH PARENTHESES TO AVOID ERRORS**

In [33]:
```
# it will print all the rows with "Date" before 2015-02-04 and "type" == "organic"
avocado[(avocado["Date"]<"2015-02-04") & (avocado["type"]=="organic")]
```

Out[33]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	regi
9173	47	2015-02-01	1.83	1228.51	33.12	99.36	0.0	1096.03	1096.03	0.00	0.0	organic	2015	Alba
9174	48	2015-01-25	1.89	1115.89	14.87	148.72	0.0	952.30	952.30	0.00	0.0	organic	2015	Alba
9175	49	2015-01-18	1.93	1118.47	8.02	178.78	0.0	931.67	931.67	0.00	0.0	organic	2015	Alba
9176	50	2015-01-11	1.77	1182.56	39.00	305.12	0.0	838.44	838.44	0.00	0.0	organic	2015	Alba
9177	51	2015-01-04	1.79	1373.95	57.42	153.88	0.0	1162.65	1162.65	0.00	0.0	organic	2015	Alba
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
11928	46	2015-02-01	1.77	7210.19	1634.42	3012.44	0.0	2563.33	2563.33	0.00	0.0	organic	2015	WestTexNewMexi
11929	47	2015-01-25	1.63	7324.06	1934.46	3032.72	0.0	2356.88	2320.00	36.88	0.0	organic	2015	WestTexNewMexi
11930	48	2015-01-18	1.71	5508.20	1793.64	2078.72	0.0	1635.84	1620.00	15.84	0.0	organic	2015	WestTexNewMexi
11931	49	2015-01-11	1.69	6861.73	1822.28	2377.54	0.0	2661.91	2656.66	5.25	0.0	organic	2015	WestTexNewMexi
11932	50	2015-01-04	1.64	6182.81	1561.30	2958.17	0.0	1663.34	1663.34	0.00	0.0	organic	2015	WestTexNewMexi

270 rows × 14 columns

Subsetting using .isin()

isin() method helps in selecting rows with having a particular(or Multiple) value in a particular column

> Syntax: DataFrame.isin(values)
>
> Parameters: values: iterable, Series, List, Tuple, DataFrame or dictionary to check in the caller Series/Data Frame.
>
> Return Type: DataFrame of Boolean of Dimension.

In [34]:
```
# subset the avocado in the region Boston or SanDiego
regionFilter = avocado["region"].isin(["Boston", "SanDiego"])
avocado[regionFilter]
```

Out[34]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year
208	0	2015-12-27	1.13	450816.39	3886.27	346964.70	13952.56	86012.86	85913.60	99.26	0.0	conventional	2015
209	1	2015-12-20	1.07	489802.88	4912.37	390100.99	5887.72	88901.80	88768.47	133.33	0.0	conventional	2015
210	2	2015-12-13	1.01	549945.76	4641.02	455362.38	219.40	89722.96	89523.38	199.58	0.0	conventional	2015
211	3	2015-12-06	1.02	488679.31	5126.32	407520.22	142.99	75889.78	75666.22	223.56	0.0	conventional	2015
212	4	2015-11-29	1.19	350559.81	3609.25	272719.08	105.86	74125.62	73864.52	261.10	0.0	conventional	2015
...	...	...	...	...	...	...	...	...	...	...	...	...	...
18100	7	2018-02-04	1.81	17454.74	1158.41	7388.27	0.00	8908.06	8908.06	0.00	0.0	organic	2018
18101	8	2018-01-28	1.91	17579.47	1145.64	8284.41	0.00	8149.42	8149.42	0.00	0.0	organic	2018
18102	9	2018-01-21	1.95	18676.37	1088.49	9282.37	0.00	8305.51	8305.51	0.00	0.0	organic	2018
18103	10	2018-01-14	1.81	21770.02	3285.98	14338.52	0.00	4145.52	4145.52	0.00	0.0	organic	2018
18104	11	2018-01-07	2.06	16746.82	5150.82	9366.31	0.00	2229.69	2229.69	0.00	0.0	organic	2018

676 rows × 14 columns

Multiple parameter Filtering

Use logical operators to combine different filters

In [36]:
```
# subset the avocado in the region Boston or SanDiego in the year 2016 or 2017
regionFilter = avocado["region"].isin(["Boston", "SanDiego"])
yearFilter = avocado["year"].isin([2016, 2017])

filtered_avocado = avocado[regionFilter & yearFilter]
filtered_avocado
```

Out[36]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year
3016	0	2016-12-25	1.28	447600.75	4349.63	346516.32	4183.69	92551.11	91481.59	1069.52	0.00	conventional	2016
3017	1	2016-12-18	1.09	579577.33	6123.84	488107.01	7765.43	77581.05	76135.49	1445.56	0.00	conventional	2016
3018	2	2016-12-11	1.22	510800.58	3711.20	409645.98	5052.84	92390.56	90449.44	1634.18	306.94	conventional	2016
3019	3	2016-12-04	1.26	473428.36	4371.95	393748.18	3449.16	71859.07	71377.77	307.69	173.61	conventional	2016
3020	4	2016-11-27	1.45	391257.01	4243.20	317090.39	3069.37	66854.05	66399.33	31.11	423.61	conventional	2016
...	...	...	...	...	...	...	...	...	...	...	...	...	...
16962	48	2017-01-29	1.21	18191.46	1477.75	8949.53	4.86	7759.32	3304.61	4454.71	0.00	organic	2017
16963	49	2017-01-22	1.73	10842.77	2019.23	6869.87	0.00	1953.67	626.78	1326.89	0.00	organic	2017
16964	50	2017-01-15	1.82	11578.42	2529.20	7637.66	0.00	1411.56	993.41	418.15	0.00	organic	2017
16965	51	2017-01-08	1.52	16775.97	2363.28	9429.06	0.00	4983.63	3266.31	1717.32	0.00	organic	2017
16966	52	2017-01-01	1.45	15752.25	1385.18	8618.28	0.00	5748.79	957.31	4791.48	0.00	organic	2017

420 rows × 14 columns

Detecting missing values .isna()

.isna() is a method used to find is there exist any NaN values in the DataFrame

It will give a True bool value if a cell has a NaN value

In [37]:
```
avocado.isna()
```

Out[37]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
18244	False	False	False	False	False	False	False	False	False	False	False	False	False	False
18245	False	False	False	False	False	False	False	False	False	False	False	False	False	False
18246	False	False	False	False	False	False	False	False	False	False	False	False	False	False
18247	False	False	False	False	False	False	False	False	False	False	False	False	False	False
18248	False	False	False	False	False	False	False	False	False	False	False	False	False	False

18249 rows × 14 columns

We can use .any() function to get a consise info

In [38]:
```
avocado.isna().any()
```

Out[38]:
```
Unnamed: 0      False
Date            False
AveragePrice    False
Total Volume    False
4046            False
4225            False
4770            False
Total Bags      False
Small Bags      False
Large Bags      False
XLarge Bags     False
type            False
year            False
region          False
dtype: bool
```

Counting missing values

In [39]:
```
avocado.isna().sum()
```

Out[39]:
```
Unnamed: 0      0
Date            0
AveragePrice    0
Total Volume    0
4046            0
4225            0
4770            0
Total Bags      0
Small Bags      0
Large Bags      0
XLarge Bags     0
type            0
year            0
region          0
dtype: int64
```

Removing missing values

- Drop NaN ** .dropna() **
- Fill NaN with value x ** .fillna(x) **

In [40]:
```
# Luckily we don't have any NaN but if we have we can use any of the two methods

avocado.dropna()

# ****  OR  ****

meanVal = avocado["AveragePrice"].mean()
avocado.fillna(meanVal)
```

Out[40]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	organic	2018	We
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	organic	2018	We
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	We
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	We
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	We

18249 rows × 14 columns

Adding a new column

It can easily be done using the [ ] brackets

Lets add a new column to our dataframe called AveragePricePer100

In [41]:
```
avocado["AveragePricePer100"] = avocado["AveragePrice"] * 100
avocado
```

Out[41]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	organic	2018	We
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	organic	2018	We
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	We
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	We
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	We

18249 rows × 15 columns

Deleting columns in DataFrame .drop(lst,axis = 1)

> dataFrame.drop(['COLUMN_NAME'], axis = 1)

- the first parameter is a list of columns to be deleted
- axis = 1 means delete column
- axis = 0 means delete row

In [42]:
```
avocado.drop(["AveragePricePer100"],axis = 1)
```

Out[42]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	organic	2018	We
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	organic	2018	We
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	We
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	We
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	We

18249 rows × 14 columns

Summary statistics

Some of the functions availabe in pandas are:

> .median() .mode() .min() .max() .var() .std() .sum() .quantile()

In [43]:
```
# mean of the AveragePrice of avocado
avocado["AveragePrice"].mean()
```

Out[43]:
```
np.float64(1.405978409775878)
```

Summarizing dates

To find the min or max date in a dataframe

In [44]:
```
avocado["Date"].max()
```

Out[44]:
```
'2018-03-25'
```

.agg() method

Pandas Series.agg() is used to pass a function or list of function to be applied on a series or even each element of series separately.

> Syntax: Series.agg(func, axis=0)
>
> Parameters: func: Function, list of function or string of function name to be called on Series. axis:0 or 'index' for row wise operation and 1 or 'columns' for column wise operation.
>
> Return Type: The return type depends on return type of function passed as parameter.

In [45]:
```
def pct30(column):
    #return the 0.3 quartile
    return column.quantile(0.3)
def pct50(column):
```

```
    #return the 0.5 quartile
    return column.quantile(0.5)

avocado[["AveragePrice","Total Bags"]].agg([pct30,pct50])
```

Out[45]:

	AveragePrice	Total Bags
pct30	1.15	7316.634
pct50	1.37	39743.830

Dropping duplicate names .drop_duplicates(lst)

Delete all the duplicate names from the dataframe

In [46]:

```
temp = avocado.drop_duplicates(subset=["year"])
temp
```

Out[46]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	re
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.00	conventional	2015	Al
2808	0	2016-12-25	1.52	73341.73	3202.39	58280.33	426.92	11432.09	11017.32	411.83	2.94	conventional	2016	Al
5616	0	2017-12-31	1.47	113514.42	2622.70	101135.53	20.25	9735.94	5556.98	4178.96	0.00	conventional	2017	Al
8478	0	2018-03-25	1.57	149396.50	16361.69	109045.03	65.45	23924.33	19273.80	4270.53	380.00	conventional	2018	Al

Count categorical data .value_counts()

Pandas Series.value_counts() function return a Series containing counts of unique values.

> Syntax: Series.value_counts(normalize=False, sort=True, ascending=False, bins=None, dropna=True)
>
> Parameter : normalize : If True then the object returned will contain the relative frequencies of the unique values. sort : Sort by values. ascending : Sort in ascending order. bins : Rather than count values, group them into half-open bins, a convenience for pd.cut, only works with numeric data. dropna : Don't include counts of NaN.
>
> Returns : counts : Series

In [47]:
```
# count number of avocado in each year in descending order
avocado["year"].value_counts(sort=True, ascending = False)
```

Out[47]:
```
year
2017    5722
2016    5616
2015    5615
2018    1296
Name: count, dtype: int64
```

Grouped summaries .groupby(col)

This function will group similar categories into one and then we can perform some summary statistics

> Syntax: DataFrame.groupby(by=None, axis=0, level=None, as_index=True, sort=True, group_keys=True, squeeze=False, **kwargs)
>
> Parameters : by : mapping, function, str, or iterable
> axis : int, default 0
> level : If the axis is a MultiIndex (hierarchical), group by a particular level or levels
> as_index : For aggregated output, return object with group labels as the index. Only relevant for DataFrame input.

> as_index=False is effectively “SQL-style” grouped output
> sort : Sort group keys. Get better performance by turning this off. Note this does not influence the order of observations within each group. groupby preserves the order of rows within each group.
> group_keys : When calling apply, add group keys to index to identify pieces
> squeeze : Reduce the dimensionality of the return type if possible, otherwise return a consistent type
>
> Returns : GroupBy object

In [48]:
```
# group by multiple columns and perform multiple summary statistic operations
avocado.groupby(["year","type"])["AveragePrice"].agg([min,max,np.mean,np.median])
```

```
C:\Windows\Temp\ipykernel_31292\3377443975.py:2: FutureWarning: The provided callable <built-in function min> is currently usin
g SeriesGroupBy.min. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass
the string "min" instead.
  avocado.groupby(["year","type"])["AveragePrice"].agg([min,max,np.mean,np.median])
C:\Windows\Temp\ipykernel_31292\3377443975.py:2: FutureWarning: The provided callable <built-in function max> is currently usin
g SeriesGroupBy.max. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass
the string "max" instead.
  avocado.groupby(["year","type"])["AveragePrice"].agg([min,max,np.mean,np.median])
C:\Windows\Temp\ipykernel_31292\3377443975.py:2: FutureWarning: The provided callable <function mean at 0x000001EDD88D6DE0> is
currently using SeriesGroupBy.mean. In a future version of pandas, the provided callable will be used directly. To keep current
behavior pass the string "mean" instead.
  avocado.groupby(["year","type"])["AveragePrice"].agg([min,max,np.mean,np.median])
C:\Windows\Temp\ipykernel_31292\3377443975.py:2: FutureWarning: The provided callable <function median at 0x000001EDD8A14680> i
s currently using SeriesGroupBy.median. In a future version of pandas, the provided callable will be used directly. To keep cur
rent behavior pass the string "median" instead.
  avocado.groupby(["year","type"])["AveragePrice"].agg([min,max,np.mean,np.median])
```

Out[48]:

year	type	min	max	mean	median
2015	conventional	0.49	1.59	1.077963	1.08
	organic	0.81	2.79	1.673324	1.67
2016	conventional	0.51	2.20	1.105595	1.08
	organic	0.58	3.25	1.571684	1.53
2017	conventional	0.46	2.22	1.294888	1.30
	organic	0.44	3.17	1.735521	1.72
2018	conventional	0.56	1.74	1.127886	1.14
	organic	1.01	2.30	1.567176	1.55

Pivot table

A pivot table is a table of statistics that summarizes the data of a more extensive table.

IMPORRANT parements to remember are
"index": it is the value that appeares on the left most side of the table (it can be a list)
"columns": these are the column you want to add to the pivot table
"aggfunc": it will call the function (it can be a list)
"values": it is the attribute which will be summarized in the table (values inside the table)

> Syntax
> pandas.pivot_table(data, values=None, index=None, columns=None, aggfunc='mean', fill_value=None, margins=False, dropna=True, margins_name='All')

> Parameters:
> data : DataFrame
> values : column to aggregate, optional
> index: column, Grouper, array, or list of the previousv columns: column, Grouper, array, or list of the previous
>
> aggfunc: function, list of functions, dict, default numpy.mean
> ....If list of functions passed, the resulting pivot table will have hierarchical columns whose top level are the function names.
> ....If dict is passed, the key is column to aggregate and value is function or list of functions
> fill_value[scalar, default None] : Value to replace missing values with
> margins[boolean, default False] : Add all row / columns (e.g. for subtotal / grand totals)
> dropna[boolean, default True] : Do not include columns whose entries are all NaN
> margins_name[string, default 'All'] : Name of the row / column that will contain the totals when margins is True.
>
> Returns: DataFrame

In [49]:

```python
# this is the same table we build in the previous cell but using pivot table
avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
```

```
C:\Windows\Temp\ipykernel_31292\762502195.py:2: FutureWarning: The provided callable <built-in function min> is currently using DataFrameGroupBy.min. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "min" instead.
  avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
C:\Windows\Temp\ipykernel_31292\762502195.py:2: FutureWarning: The provided callable <built-in function max> is currently using DataFrameGroupBy.max. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "max" instead.
  avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
C:\Windows\Temp\ipykernel_31292\762502195.py:2: FutureWarning: The provided callable <function mean at 0x000001EDD88D6DE0> is currently using DataFrameGroupBy.mean. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "mean" instead.
  avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
C:\Windows\Temp\ipykernel_31292\762502195.py:2: FutureWarning: The provided callable <function median at 0x000001EDD8A14680> is currently using DataFrameGroupBy.median. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "median" instead.
  avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
```

Out[49]:

		min	max	mean	median
		AveragePrice	AveragePrice	AveragePrice	AveragePrice
year	**type**				
2015	**conventional**	0.49	1.59	1.077963	1.08
	organic	0.81	2.79	1.673324	1.67
2016	**conventional**	0.51	2.20	1.105595	1.08
	organic	0.58	3.25	1.571684	1.53
2017	**conventional**	0.46	2.22	1.294888	1.30
	organic	0.44	3.17	1.735521	1.72
2018	**conventional**	0.56	1.74	1.127886	1.14
	organic	1.01	2.30	1.567176	1.55

Explicit indexes

Indexes make subsetting simpler using .loc and .iloc

In [50]:

```
regionIndex = avocado.set_index(["region"])
regionIndex
```

Out[50]:

region	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	ty
Albany	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventio
Albany	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventio
Albany	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventio
Albany	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventio
Albany	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventio
...	...	...	...	...	...	...	...	...	...	...	...	
WestTexNewMexico	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	orga
WestTexNewMexico	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	orga
WestTexNewMexico	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	orga
WestTexNewMexico	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	orga
WestTexNewMexico	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	orga

18249 rows × 14 columns

In [51]:

```
# Insted of doing this
```

```python
avocado[avocado["region"].isin(["Albany", "WestTexNewMexico"])]
```

Out[51]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	organic	2018	We
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	organic	2018	We
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	We
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	We
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	We

673 rows × 15 columns

In [52]:
```python
# we can simply do
```

```
regionIndex.loc[["Albany", "WestTexNewMexico"]]
```

Out[52]:

region	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	ty
Albany	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventio
Albany	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventio
Albany	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventio
Albany	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventio
Albany	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventio
...	...	...	...	...	...	...	...	...	...	...	...	
WestTexNewMexico	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13066.82	431.85	0.0	orga
WestTexNewMexico	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940.04	324.80	0.0	orga
WestTexNewMexico	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	orga
WestTexNewMexico	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	orga
WestTexNewMexico	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	orga

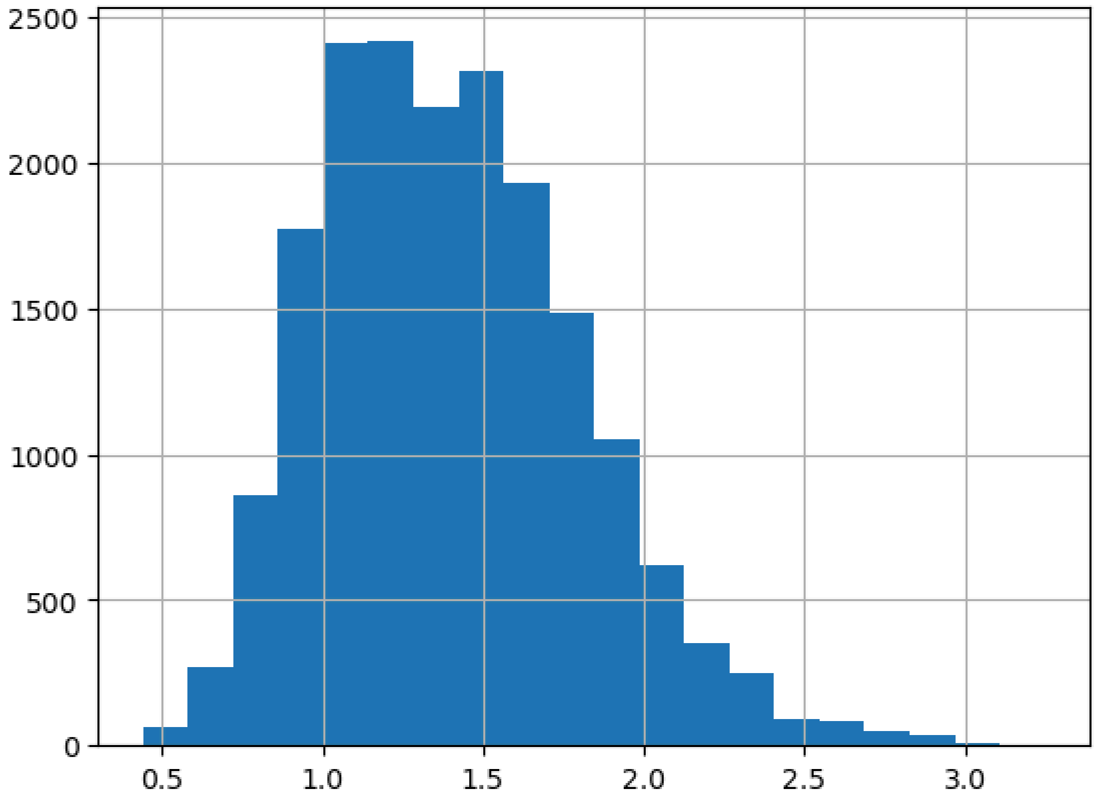
673 rows × 14 columns

Visualizing your data

Histograms

use the function .hist()

In [53]:
```
avocado["AveragePrice"].hist(bins=20)
plt.show()
```



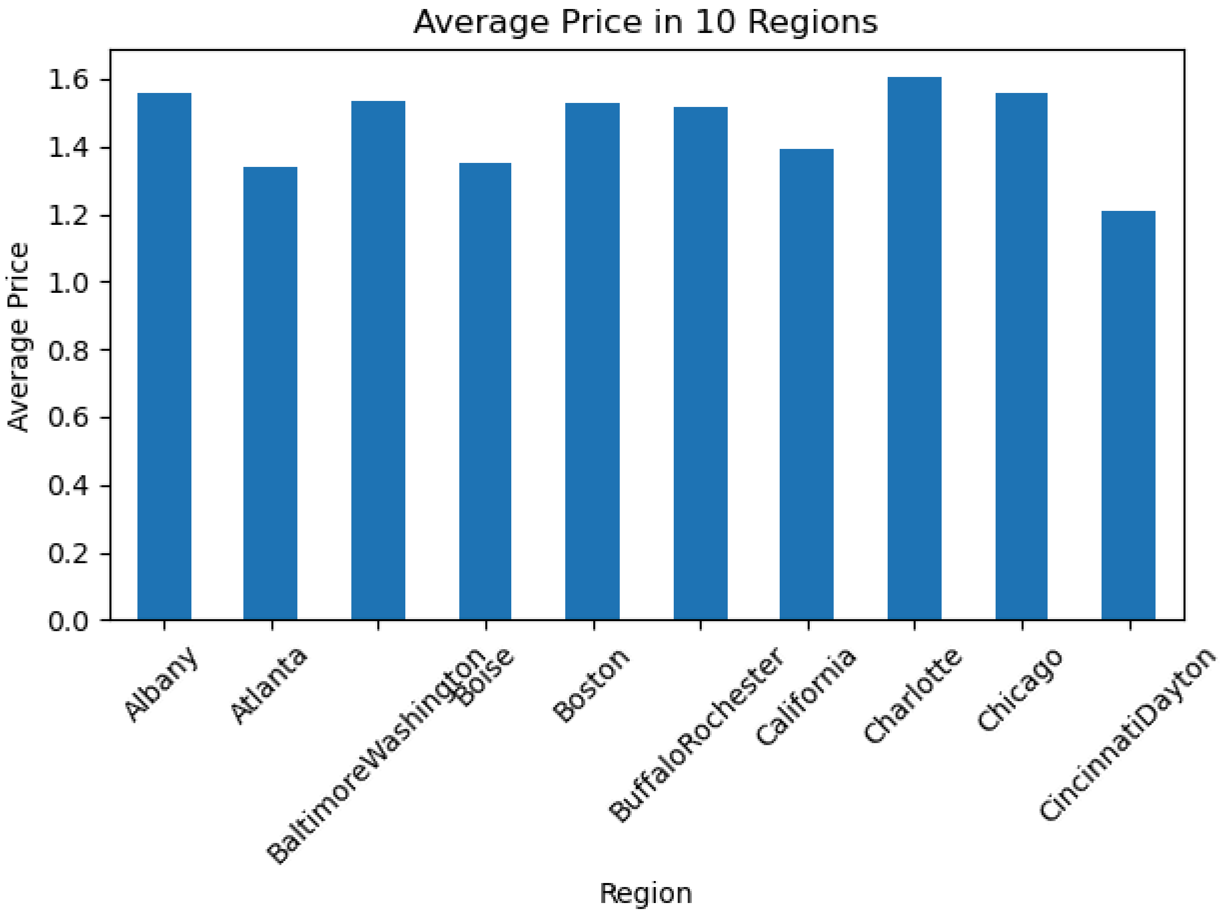
Bar plots

In [54]:
```
regionFilter = avocado.groupby("region")["AveragePrice"].mean().head(10)
regionFilter
```

Out[54]:
```
region
Albany                 1.561036
Atlanta                1.337959
BaltimoreWashington    1.534231
Boise                  1.348136
Boston                 1.530888
BuffaloRochester       1.516834
California             1.395325
Charlotte              1.606036
Chicago                1.556775
CincinnatiDayton       1.209201
Name: AveragePrice, dtype: float64
```

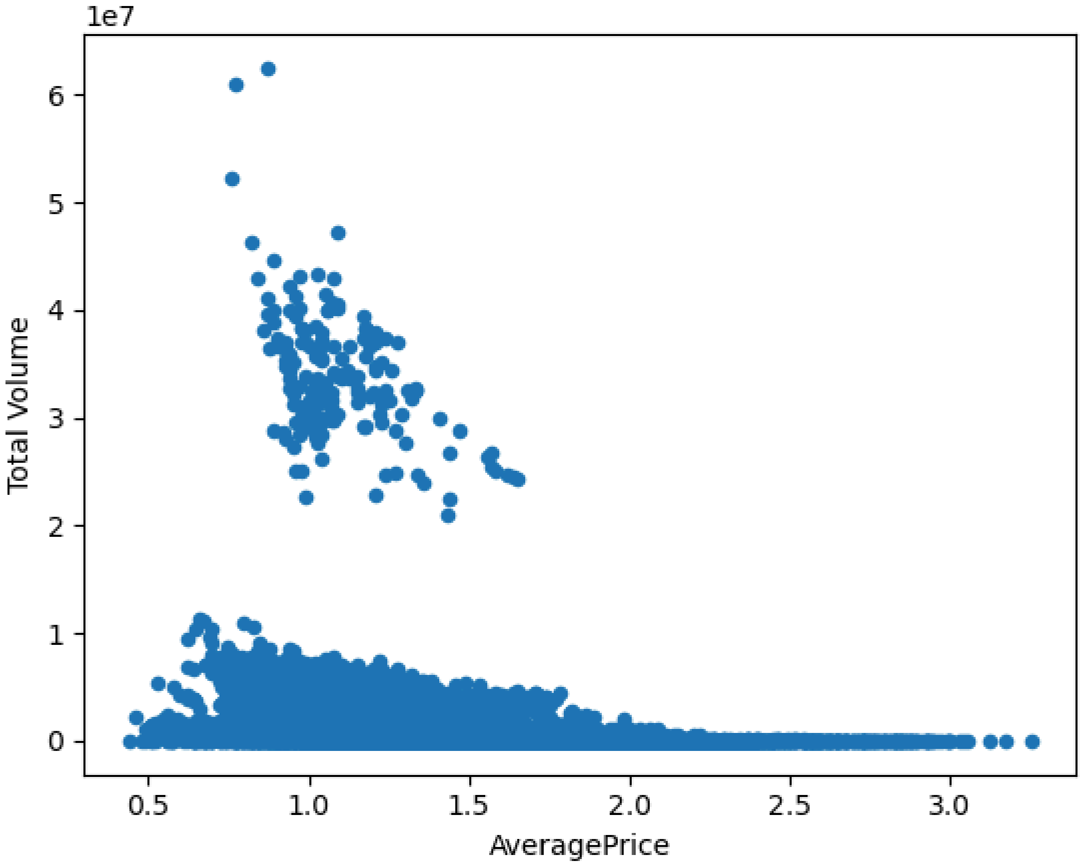
In [56]:
```
avocado.groupby("region")["AveragePrice"].mean().head(10).plot(
    kind="bar",
    rot=45,
    title="Average Price in 10 Regions"
)

plt.xlabel("Region")
plt.ylabel("Average Price")
plt.tight_layout()
plt.show()
```



Scatter plot

In [59]:
```
avocado.plot(x="AveragePrice", y="Total Volume", kind="scatter")
plt.show()
```



Arithmetic with Series & DataFrames

You can use arithmetic operators directly on series but sometimes you need more control while performing these operations, here is where these explicit arithmetic functions come into the picture

Add/Subtract function (just replece add with sub)

```
Syntax: Series.add(other, level=None, fill_value=None, axis=0)
```

```
Parameters:
other: other series or list type to be added into caller series
fill_value: Value to be replaced by NaN in series/list before adding
level: integer value of level in case of multi index

Return type: Caller series with added values
```

Multiplication function

```
Syntax: Series.mul(other, level=None, fill_value=None, axis=0)

Parameters:
other: other series or list type to be added into caller series
fill_value: Value to be replaced by NaN in series/list before adding
level: integer value of level in case of multi index

Return type: Caller series with added values
```

Division function

```
Syntax: Series.div(other, level=None, fill_value=None, axis=0)

Parameters:
other: other series or list type to be divided by the caller series
fill_value: Value to be replaced by NaN in series/list before division
level: integer value of level in case of multi index

Return type: Caller series with divided values
```

In [60]:

```
# subtract AveragePrice with AveragePrice :P
# Dah its 0
avocado["AveragePrice"].sub(avocado["AveragePrice"])
```

Out[60]:
```
0        0.0
1        0.0
2        0.0
3        0.0
4        0.0
        ...
18244    0.0
18245    0.0
18246    0.0
18247    0.0
18248    0.0
Name: AveragePrice, Length: 18249, dtype: float64
```

Merge DataFrames

Syntax:

> DataFrame.merge(self, right, how='inner', on=None, left_on=None, right_on=None, left_index=False, right_index=False, sort=False, suffixes=('_x', '_y'), copy=True, indicator=False, validate=None) → 'DataFrame'[source]¶ Merge DataFrame or named Series objects with a database-style join.

The join is done on columns or indexes. If joining columns on columns, the DataFrame indexes will be ignored. Otherwise if joining indexes on indexes or indexes on a column or columns, the index will be passed on.

Parameters right: DataFrame or named Series Object to merge with.

how{'left', 'right', 'outer', 'inner'}, default 'inner'

on: label or list Column or index level names to join on. These must be found in both DataFrames. If on is None and not merging on indexes then this defaults to the intersection of the columns in both DataFrames.

left_on: label or list, or array-like Column or index level names to join on in the left DataFrame. Can also be an array or list of arrays of the length of the left DataFrame. These arrays are treated as if they are columns.

right_on: label or list, or array-like Column or index level names to join on in the right DataFrame. Can also be an array or list of arrays of the length of the right DataFrame. These arrays are treated as if they are columns.

left_index: bool, default False Use the index from the left DataFrame as the join key(s). If it is a MultiIndex, the number of keys in the other DataFrame (either the index or a number of columns) must match the number of levels.

right_index: bool, default False Use the index from the right DataFrame as the join key. Same caveats as left_index.

sort: bool, default False Sort the join keys lexicographically in the result DataFrame. If False, the order of the join keys depends on the join type (how keyword).

suffixes: tuple of (str, str), default (‘_x’, ‘_y’) Suffix to apply to overlapping column names in the left and right side, respectively. To raise an exception on overlapping columns use (False, False).

Join

> DataFrame.merge(self, right, how='inner', on=None, left_on=None, right_on=None, left_index=False, right_index=False, sort=False, suffixes=('_x', '_y'), copy=True, indicator=False, validate=None) → 'DataFrame'[source]¶ Merge DataFrame or named Series objects with a database-style join.

The join is done on columns or indexes. If joining columns on columns, the DataFrame indexes will be ignored. Otherwise if joining indexes on indexes or indexes on a column or columns, the index will be passed on.

Parameters rightDataFrame or named Series Object to merge with.

how{‘left’, ‘right’, ‘outer’, ‘inner’}, default ‘inner’ on: label or list Column or index level names to join on. These must be found in both DataFrames. If on is None and not merging on indexes then this defaults to the intersection of the columns in both DataFrames.

left_on: label or list, or array-like Column or index level names to join on in the left DataFrame. Can also be an array or list of arrays of the length of the left DataFrame. These arrays are treated as if they are columns.

right_on: label or list, or array-like Column or index level names to join on in the right DataFrame. Can also be an array or list of arrays of the length of the right DataFrame. These arrays are treated as if they are columns.

left_index: bool, default False Use the index from the left DataFrame as the join key(s). If it is a MultiIndex, the number of keys in the other DataFrame (either the index or a number of columns) must match the number of levels.

right_index: bool, default False Use the index from the right DataFrame as the join key. Same caveats as left_index.

sort: bool, default False Sort the join keys lexicographically in the result DataFrame. If False, the order of the join keys depends on the join type (how keyword).

suffixes: tuple of (str, str), default ('_x', '_y') Suffix to apply to overlapping column names in the left and right side, respectively. To raise an exception on overlapping columns use (False, False).

What next?

Try to use your skills on some other dataset

Have a look at my analysis of some other datasets :P

- Android App Market on Google Play
- MNIST Ensemble of 5 CNNS to get 0.99742 score!!
- Titanic Survival

Please upvote if you find it helpful :D

In [ ]:

```
```